Information Visualization

Course introduction

Lesson 1

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Table of contents

01

Overview

What we talk about, what you are interested in (topic-wise) and what you want to take form this course

Data analysis

Methods to query, process, analyse Linked Data with Python

Data visualization

Methods and techniques to plot data with Python and Js

Web communication

Presentation techniques for explanatory projects

Required background

Programming

Python (intermediate)

Install libraries, Jupyter notebook, read/write CSV and JSON data

Web tech: UI / UX

HTML, CSS, JS (good)

JQuery for UI, modify DOM, interactivity

Introductory methods

GitHub (good)

A short introduction
Github guides
sourcetree GUI

Knowledge mgmt.

RDF, SPARQL, OWL (good)

Read RDF in several syntaxes, read/write SPARQL queries, understand basics of ontologies

Expected improved skills

Programming

Python and Jupyter

Libraries for data exploration.
Jupyter to document your work

Web tech: UI / UX

JS (good)

Libraries for data visualization,
Digital storytelling strategies

Introductory methods

Github, Colab

Publish your work on github (data, software and website)

Knowledge mgmt.

RDF, SPARQL, OWL (good)

Python APIs for RDF/SPARQL

Classes overview

14/4 Introduction and basic concepts
15/4 Basic concepts
21/4 RDF and SPARQL (1)
22/4 RDF and SPARQL (2)
5/5 Data sense making (1)
6/5 Data sense making (2)
12/5 Data visualisation
13/5 Storytelling
19/5 Workshop
20/5 Wrap up
21/5 Early exam session (time and place to be confirmed, AM)

02

Early exam or project?

An early exam session or a group project

Early exam session

NEW!!

If you do not fancy yet-another-group-project, you can opt for the “zero session” on **May 21 2026**.

This modality requires you to take a **written test**.

The test is individual and lasts **2 hours.**

Early exam session NEW!!

The test requires you to accomplish a number of **tasks** tailored on the skills you should have gained at that point and to describe your **choices** via open-ended questions.

Tasks: data collection - knowledge extraction - analysis - visualisation and interpretation. All topics are addressed during the course.

The focus of the evaluation is on your interpretation and **critical skills**. You can use slides, web, and genAI during the test (but you likely won't for the important things…).

*This assessment is designed so that generic or AI-generated answers will be obvious and score poorly. The strongest submissions will be **specific and personal**.*

Early exam – preparation

NEW!!

02

To prepare for the early session you MUST:

1) Read the slides, practice exercises and hands-ons
2) **Prepare a dataset with some personal data*** (you will have to upload it)
3) Choose one analysis task to perform on your data
4) **Prepare the python code*** to process your data (you will have to upload a py or ipynb file)
5) **Prepare a chart*** (can be the output of the script and/or a post-processed version, you will have to upload it as jpg or html)
6) Prepare a critical disclosure on data design choices, e.g. why, how, what for… (~500 words describing what the data is about and the added value). This is to practice potential questions, it won't be uploaded nor evaluated *- genAI not allowed here*

Early exam – preparation NEW!!

(*) What you can do ahead:

a. knowledge extraction operations on your data (read/filter/write) in python using a code editor of your choice (VScode, colab, etc) *- genAI allowed*
b. create static/dynamic visualisation (using a py/js or a tool of your choice) *- genAI allowed*
c. create a short narrative to accompany the chart *- genAI not allowed*

Early exam – data preparation*

NEW!!

Collect a dataset with **your personal data.**

The data collection must be done **(semi-)manually** (automated tracking apps should be discussed), in retrospective or via daily observations.

Examples (pick one or choose a new one):

- *Reading habits*
- *Archive, museum, or library visits*
- *Time spent with/number of different types of documents*
- *Study or writing behaviour*

Early exam – data preparation* NEW!!

Max 200 data points / rows.

Allowed formats: json, csv, txt, rdf.

Anonymise data if applicable or necessary (e.g. use pseudonyms).

The dataset should be relatively small but should encompass all information relevant to perform some sort of analysis that you believe is important (e.g. dates or durations if temporal analysis is relevant, places, names, emotions, etc.). These pieces of information will be the columns of your table (if csv) or the key value attributes (if json).

Early exam evaluation NEW!!

The evaluation of the written test is **qualitative** (open questions using a google form).

Final grades are computed based on the average score of the answers above, and are communicated in the following days via email for acceptance.

The early exam session is **not mandatory**. The date is final and there are no extra sessions.

If you fail or do not accept the grade, you'll have to do the project.

Early exam evaluation NEW!!

What I value (and evaluate)

To prevent genAI to do the important bits of your job, the exam fosters the usage of embodied knowledge, ethics, reflexivity, and design practice.

Translate your experience into structured data
Make explicit, defensible design decisions
Critically reflect on ethical implications of visualising personal data
Articulate what visualisation cannot or should not do

Failing to address these aspects leads to lower grades, regardless of the correctness of the analysis or coding aspects.

Project

02

A web project

Choose a **topic**,
find data-driven **questions**,
analyse and **visualize** data,
produce a **notebook** with your code,
and a **website** for presenting
results.

MUST CHOOSE

Topic

Pick a domain, and quantitative questions

Data

Find or create data that support your analysis

Group

Max 3-people group to share the pain

Topic: mandatory

The topic of the project must be related to **Art History** or **History of photography**. To propose a different topic, contact me with idea, questions*, and data sources already defined.

Topics should be articulated into one or more **research questions**, which in turn must be answered using quantitative data analysis and visualisation.

Once you defined your research questions, email me ***marilena.daquino2@unibo.it*** *for feedback and suggestions.*

Data: warmly recommended

The course will make use of data from **artresearch.net linked open data source.**

Consider integrating multiple sources to answer your questions. **You are warmly recommended to use at least two data sources**, out of which, at least one must be a Linked Open Dataset.

Group: 1–3 people

3 people max. Justify your contribution to the project. Grades are individual (you are judged for your contribution).

You can work alone, but you need to achieve the same results. No discounts :(

Result: a notebook, a website

All data preparation (query, merging, cleaning, etc.) must be documented in a **jupyter notebook.**

Charts and narrative are to be included in the **website.**

****Somewhere in between: you can use platforms like streamlit.io to both process data and present results in one place.****

03

Project evaluation

What to prepare, what happens the day of the exam, how I grade your work.

By the day before at noon

MUST HAVE

Presentation

25′ **presentation**
(no slides)

Jupyter notebook

A **jupyter** notebook

Website

An **online webpage**

By the day before at noon

Open an issue on the course repo

Open or comment an issue on the repo of the course called “Exam DD/MM/YYYY” with: Project title, Website URL, Repository URL, People **[here]**

Or send an email

If you agreed on a different program or for any other issue, email me.

The project will be preliminarily evaluated the day before the exam *(that’s why you need to send it by noon).*

If the project is good enough, we may agree on an offline evaluation (I send you comments and grade via email). We can always have a chat during the exam session.

What to submit

Jupyter notebook

- Use Jupyter [1] or use Colab
- 1 main notebook per project
- Write a short abstract in the notebook
- Query data and process them for the visualisations
- Document functions
- Keep it short!

Website

- A static website is sufficient
- Access data prepared in the notebook or query live services
- Show selected charts accompanied by your narrative
- Include conclusions, credits (people and roles), and licenses
- The website **must be online**

Github repo

- Upload notebook, data (optional) and website (optional) on the repo

The day of the exam

HOW IT GOES

Presentation

You present your project while showing the **website** (25’)

Grading review

While you present I may review the grade I already assigned you

Q&A

Discussion of methods, **results** and questions

Can/not do

- Reuse existing **templates** for the website (e.g. CMS, HTML templates)
- Reuse any **py/JS libraries**
- Other solutions than **GitHub** to publish the website. However, a git repo is mandatory to publish the notebook.
- You must **produce original content**. You can use genAI to improve your text, but the content must be original, well written, and be precise.
- You **can** use genAI to help you in coding tasks and proofreading. That depends on how proficient you are or want to become in some technology or communication technique.
- You **cannot** use genAI to interpret charts and present results.

Evaluation

SKILLS

Data	Research		Presentation	
Correct queries	Sound questions		Clarity	26
Correct results	Adequate visuals		Summarisation	28
Data integration	Graphic skills	30	Surprise me!	30L

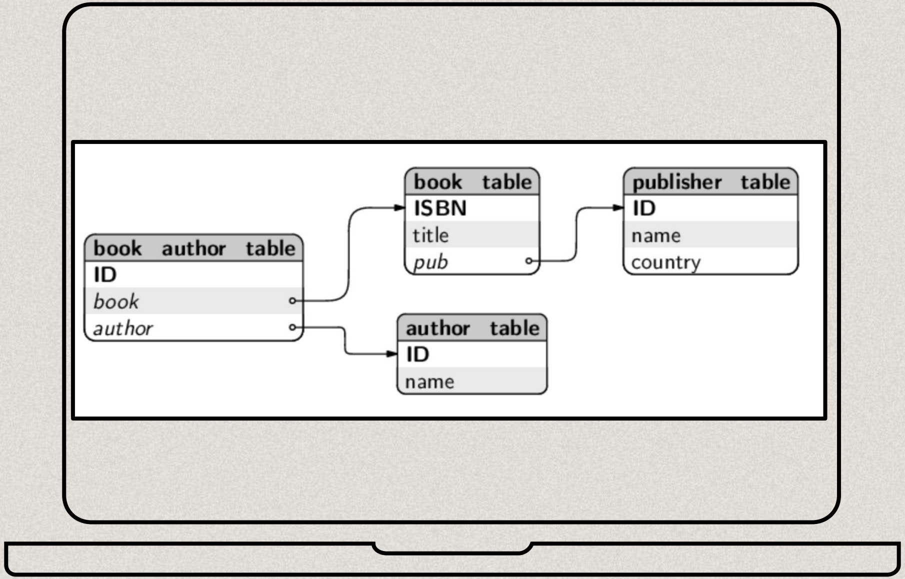
04

Basic concepts

Data and visualization dimensions

01

Data attributes



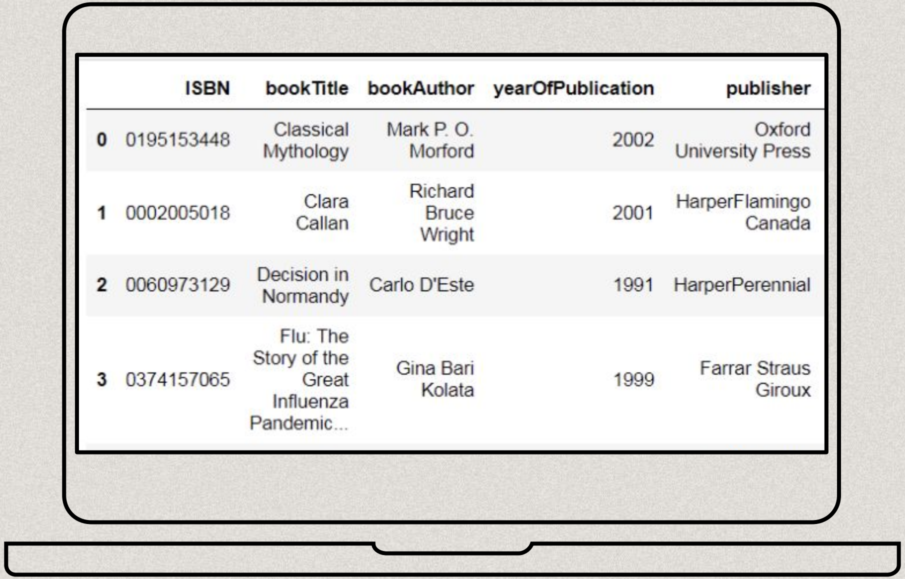
Data

Data are abstractions of concepts and real-world **entities** (people, books, places)

Data have **attributes**, also called features or variables (title, author, ISBN are attributes of books)

01

Data attributes



	ISBN	bookTitle	bookAuthor	yearOfPublication	publisher
0	0195153448	Classical Mythology	Mark P. O. Morford	2002	Oxford University Press
1	0002005018	Clara Callan	Richard Bruce Wright	2001	HarperFlamingo Canada
2	0060973129	Decision in Normandy	Carlo D'Este	1991	HarperPerennial
3	0374157065	Flu: The Story of the Great Influenza Pandemic...	Gina Bari Kolata	1999	Farrar Straus Giroux

Dataset

A dataset is a **collection** of entities with their attributes.

It can be seen as an **n*m matrix**, wherein n is the number of rows (entities) and m is the number of attributes (columns)

Attributes

Numeric

Measurable **quantities** (e.g. temperature, price)

Categorical

[also called Nominal] **Names** from non-overlapping sets, classes, or states (e.g. title, sex)

Ordinal

Nominal or numeric attributes that can be **ranked** (e.g. days, years, Likert: “strongly like” to “strongly dislike”).

Attributes

Numeric

Measurable **quantities** (e.g. temperature, price)

Interval attributes are measured on the basis of a scale with an interval and an origin (e.g. time, temperature). They have no true-zero.

Ratio attributes have a true-zero - meaning no quantity is measured in that point (e.g. price)

Operations

Numeric

Interval can be **ordered**, but not multiplied/divided.

On Ratio you can do **arithmetic** operations.

Categorical

Can be **sorted** (e.g. alphabetically) and counted.

Cannot be ordered and arithmetic operations cannot be performed.

Ordinal

Can be naturally **ordered** and counted.

Arithmetic operations are not possible.

What can you do with these data?

Answer the questionnaire (10 minutes)

https://forms.gle/caSW8W4nPaavsir56

Graphical properties

Some properties of a data visualization help making noticeable graphical elements

Axes

Layout

Shape

Colour

Size

Typography

Axes

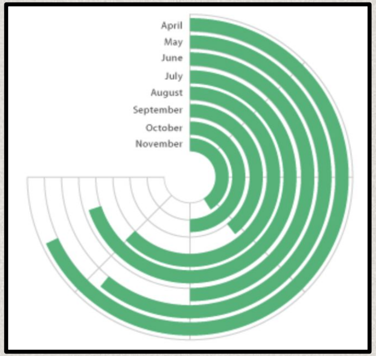
02

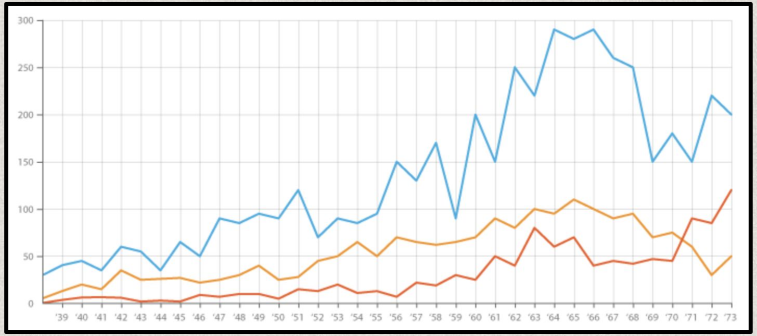
Graphical properties

Cartesian/radial axes

A visual guide for the **placement** of elements composing the visualisation.

A visualization with axes is called **graph**, otherwise it is called chart (although chart is used for any kind of visual).





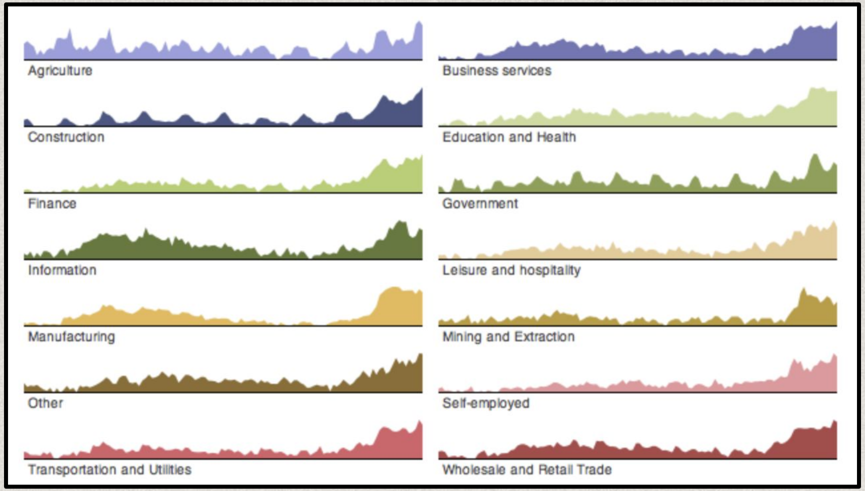
Layout

Single/Multiple canvas

The **format and symmetry** of the visualisation change according to the volume of data and the number of attributes to show.

More visualisations in the canvas help comparison, but hide the big picture!

A viz. should always fit the frame (e.g. a screen).



Shape

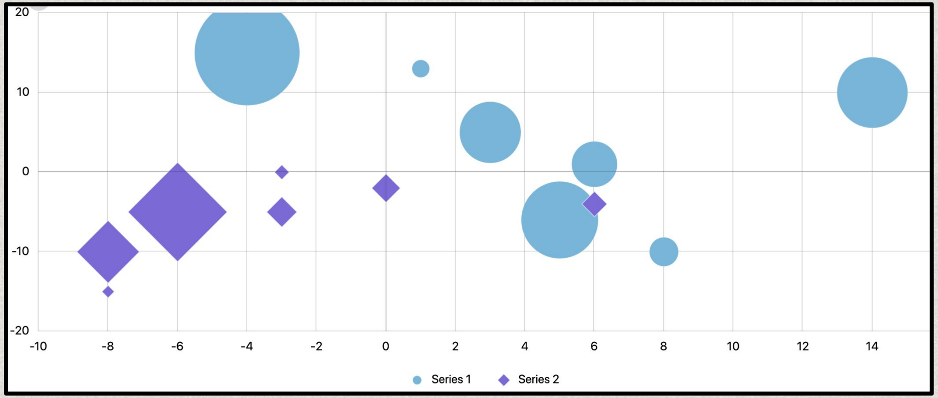
Graphical properties

Shapes and glyphs

Realistic (icons) or abstract (circles) help to **distinguish** elements.

Shapes and glyphs demand more **attention** (icons can mitigate the effort)

Lines are effective shapes (highlight a trend)



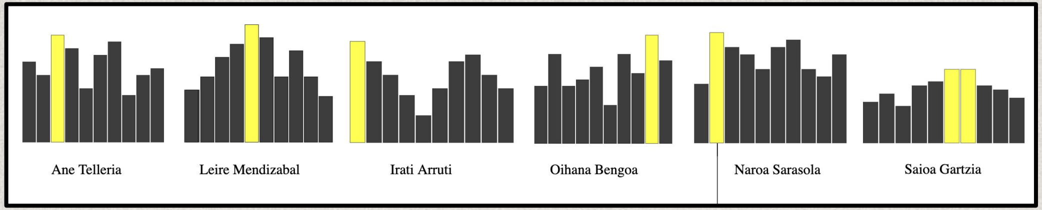
Colour

Palette and luminosity

Color differences are detected in <200 milliseconds (preattentive perception). Must be natural (e.g. found in nature).

Used to **distinguish elements or patterns** in big datasets, it’s less useful in small datasets (one color is sufficient).

Luminosity highlights relevant patterns (bright colors pop out, while dark colors recede).

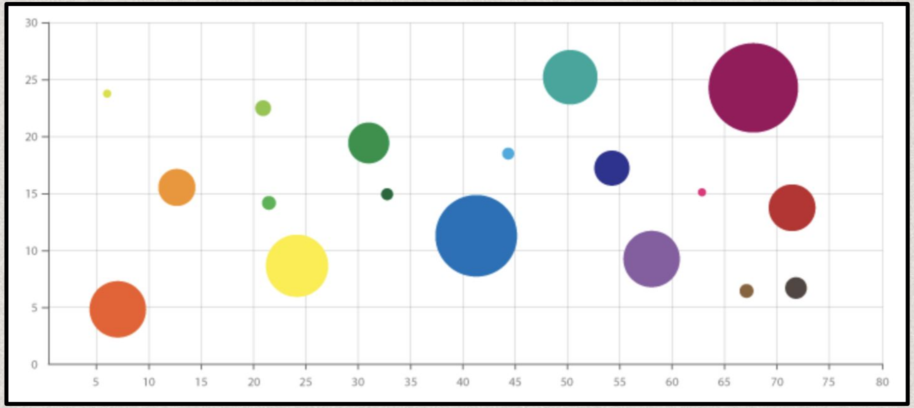


Size

Show the variety

Identify **variations between similar elements** by size (regardless of other dimensions) is the quickest way.

Still, it is not the most effective means to compare values (e.g. circle areas).



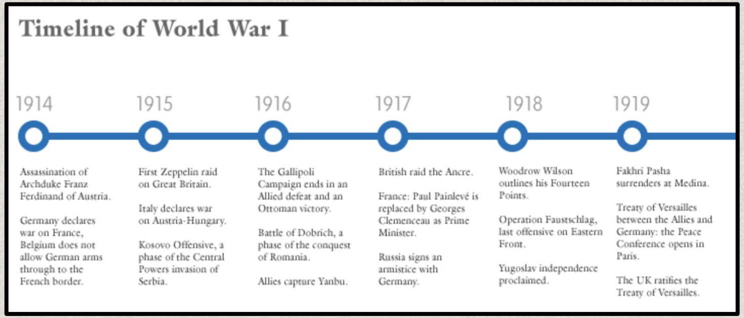
Typography

02

Graphical properties

Primary/secondary role

Can be a primary aspect (e.g. wordle) along with other dimensions (e.g. size) or secondary, to describe the visualization.



Interaction and context

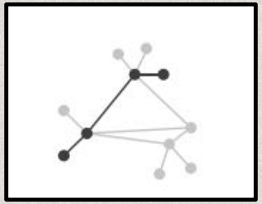
Graphical properties interact with each other to convey the message in different contexts.

Location

Placement helps to familiarise with data (e.g. a map)

Network

Closeness and **connections** of data points (e.g. a network graph)



Sequence

Axes-based vis. create a **linear** reading (e.g. line graphs)

Graphical properties + Context =

Visual pattern

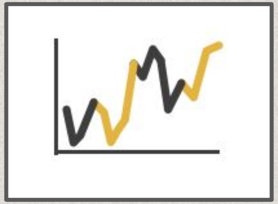
By means of graphical and contextual aspects, what should emerge from visualisations is a **visual pattern**. Several co-occurring dimensions can contribute to highlight patterns.

How do visual patterns look like

Differentiation

Contrast values


distribution


samples


similarity

How do visual patterns look like

Gradation

Continuous values

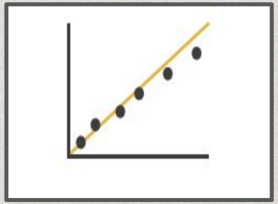
trend

How do visual patterns look like

Anomaly

Break the **pattern.**

outliers

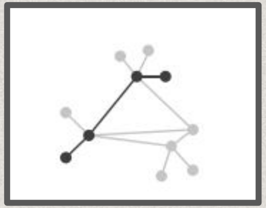


correlation

How do visual patterns look like

Paths

Position and relations



Clusters and paths

Perception

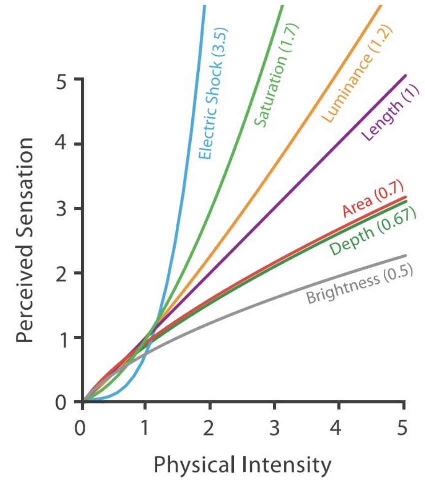
Preattentive variables

Visual aspects have a **functional** role in the interpretation of data rather than aesthetic.

The idea of visualising data in a graphical form is to **replace cognition with perception.**

The human eye perceives visual dimensions in different ways, with more or less effort, and users' **knowledge background** can affect the perception.

Nonetheless, some dimensions are more **intense** than others.



Steven's Psychophysical Power Law: $S = I^N$

Stevens, 1975

Steven's power law

Tidwell's preattentive variables

Some visual aspects work **preattentively**, meaning they are able to communicate something before the user pays conscious attention to it.

0.103	0.176	0.387	0.300	0.379	0.276	0.179	0.321	0.192	0.250
0.333	0.384	0.564	0.587	0.857	1.064	0.698	0.621	0.232	0.316
0.421	0.309	0.654	0.729	0.228	0.529	0.832	0.935	0.452	0.426
0.266	0.750	1.056	0.936	0.911	0.820	0.723	1.201	0.935	0.819
0.225	0.326	0.643	0.337	0.721	0.837	0.682	0.987	0.984	0.849
0.187	0.586	0.529	0.340	0.829	0.835	0.873	0.945	1.103	0.710
0.153	0.485	0.560	0.428	0.628	0.335	0.956	0.879	0.699	0.424

Find numbers greater than 1

Tidwell's preattentive variables

02

Preattentive variables

Effective and impactful data visualisations work extensively on these aspects.

Tidwell et al, Designing interfaces found 8 variables.

0.103	0.176	0.387	0.300	0.379	0.276	0.179	0.321	0.192	0.250
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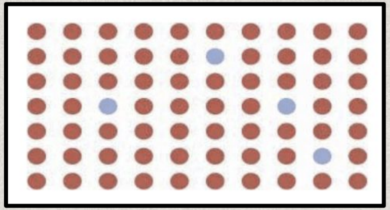
Find numbers greater than 1



0.103	0.176	0.387	0.300	0.379	0.276	0.179	0.321	0.192	0.250
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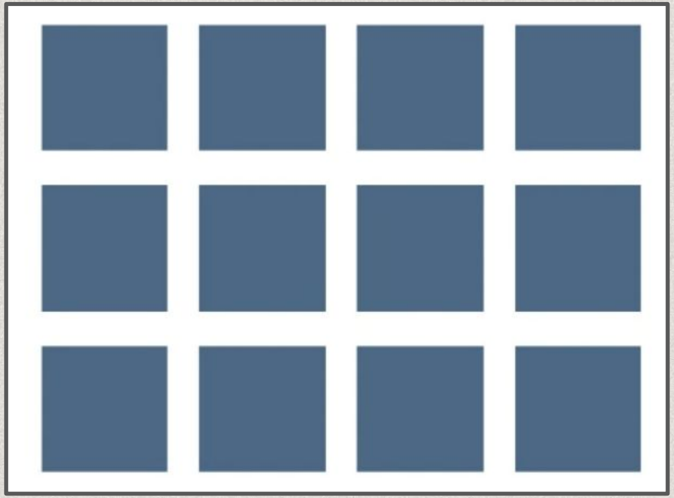
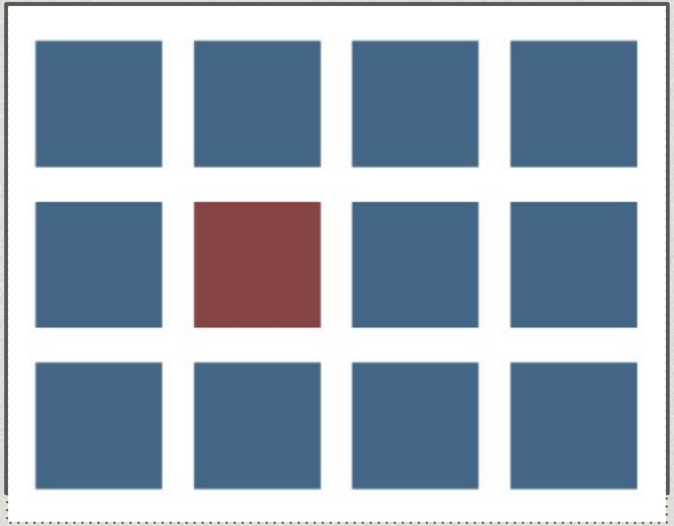
Tidwell's preattentive variables



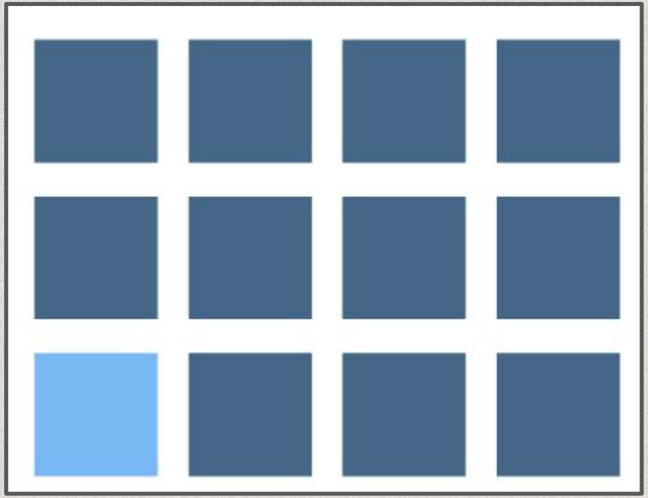
Let's start!

Tidwell's preattentive variables



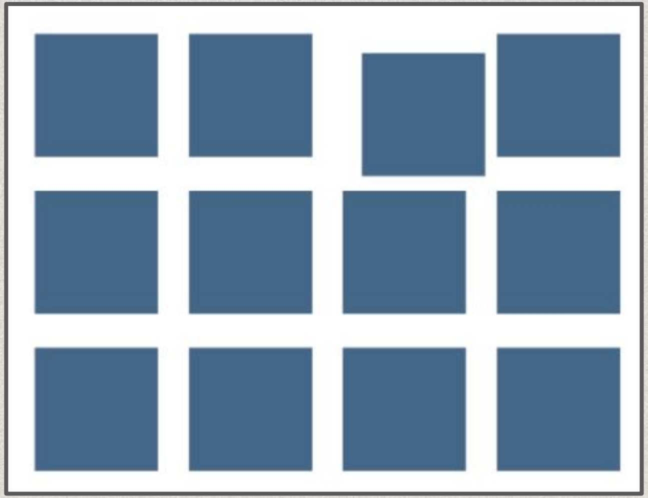
Color hue

Tidwell's preattentive variables



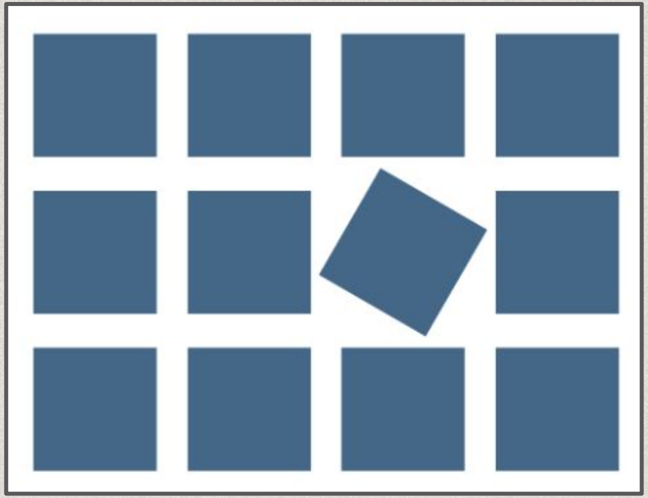
Color brightness

Tidwell's preattentive variables



Position and alignment

Tidwell's preattentive variables

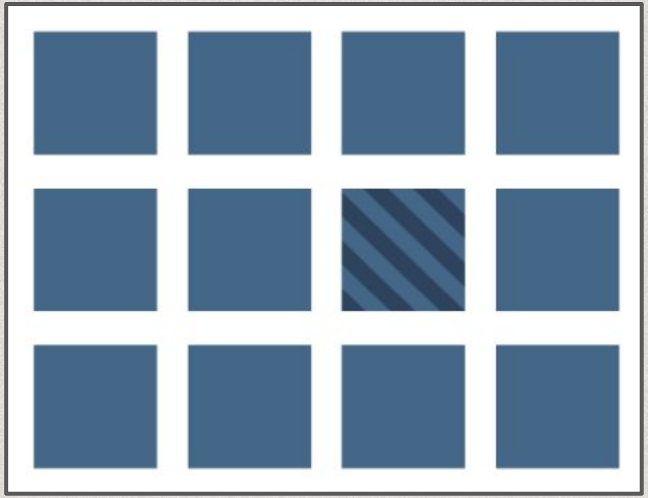


Orientation

Tidwell's preattentive variables

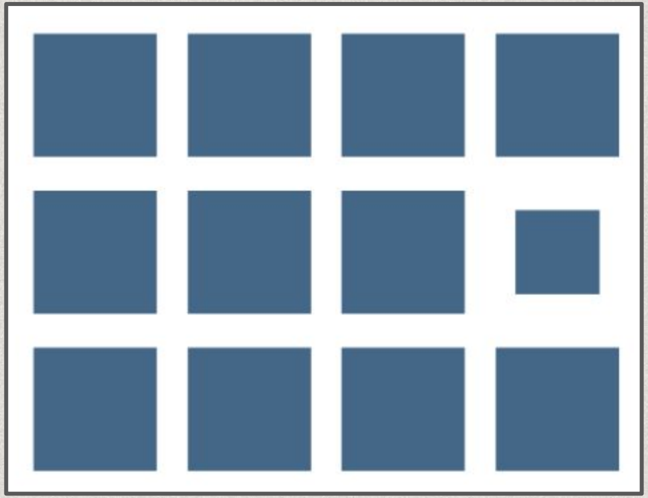
Color saturation

Tidwell's preattentive variables



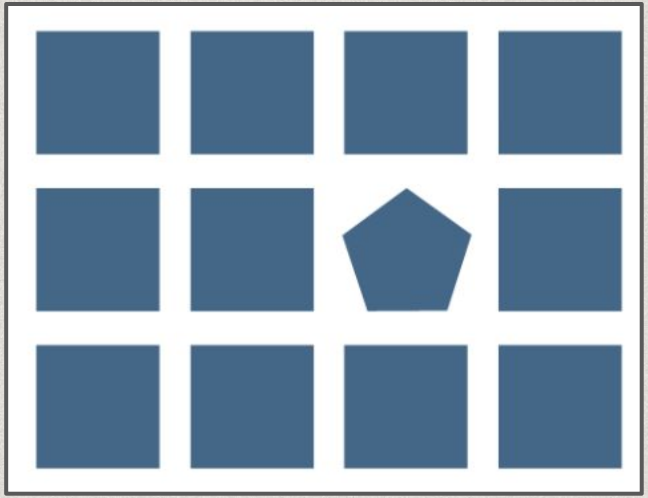
Texture

Tidwell's preattentive variables

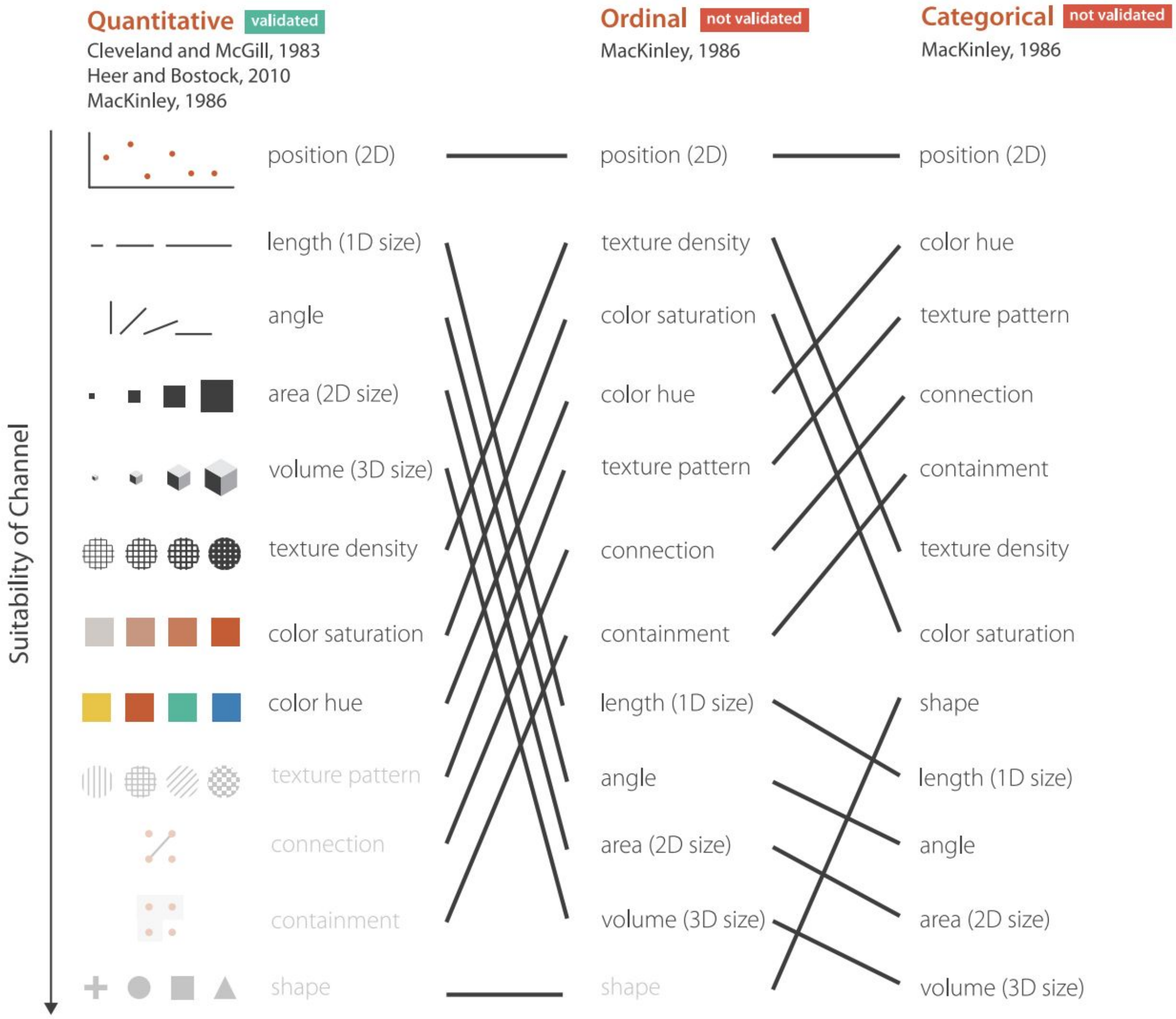


Size (or width, length)

Tidwell's preattentive variables



Shape

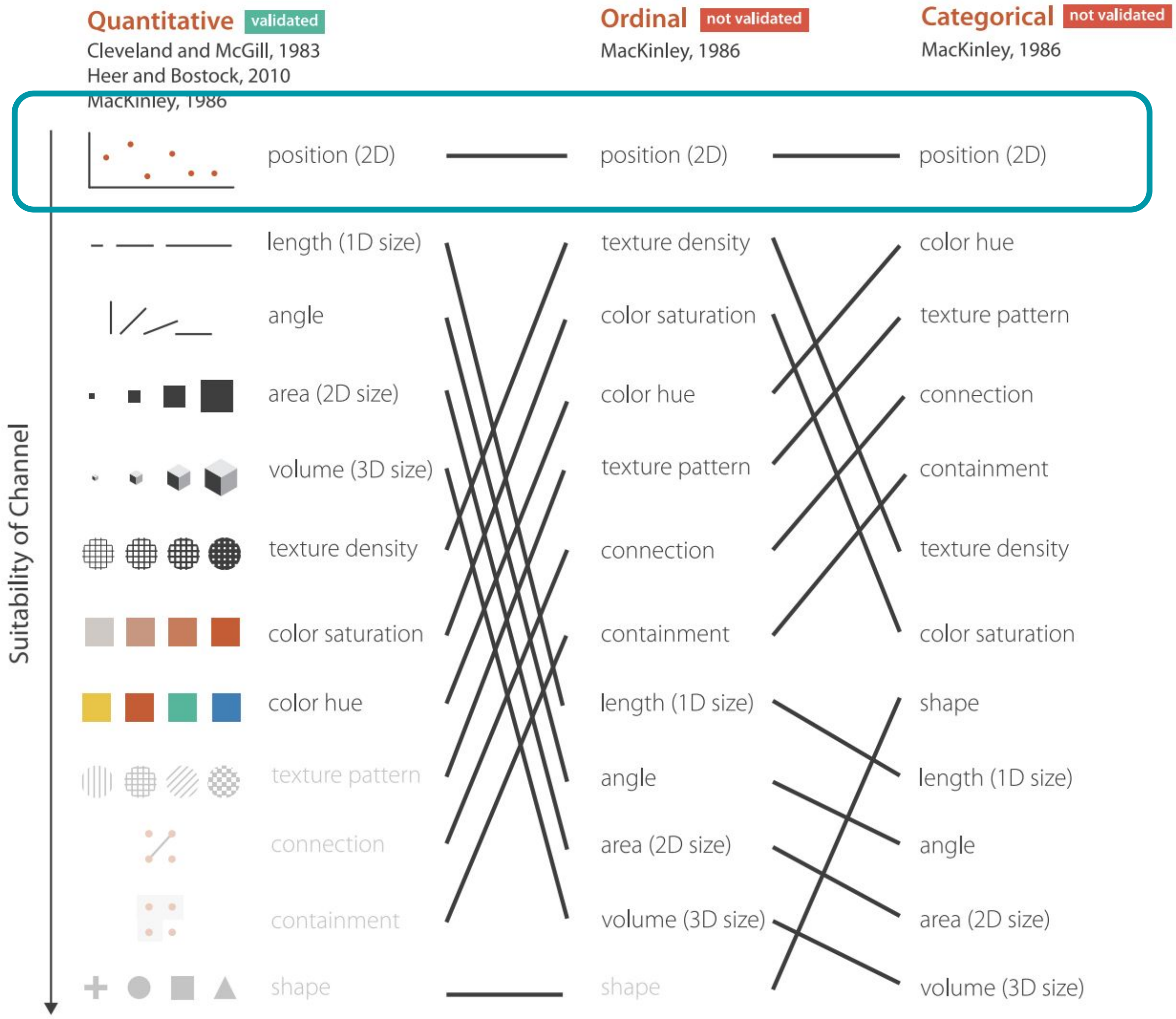


03

Best practices

Studies demonstrated that according to the **type of data attributes** at hand, certain **graphical properties** work better than others.

In particular, properties are perceived with more or less **accuracy**.

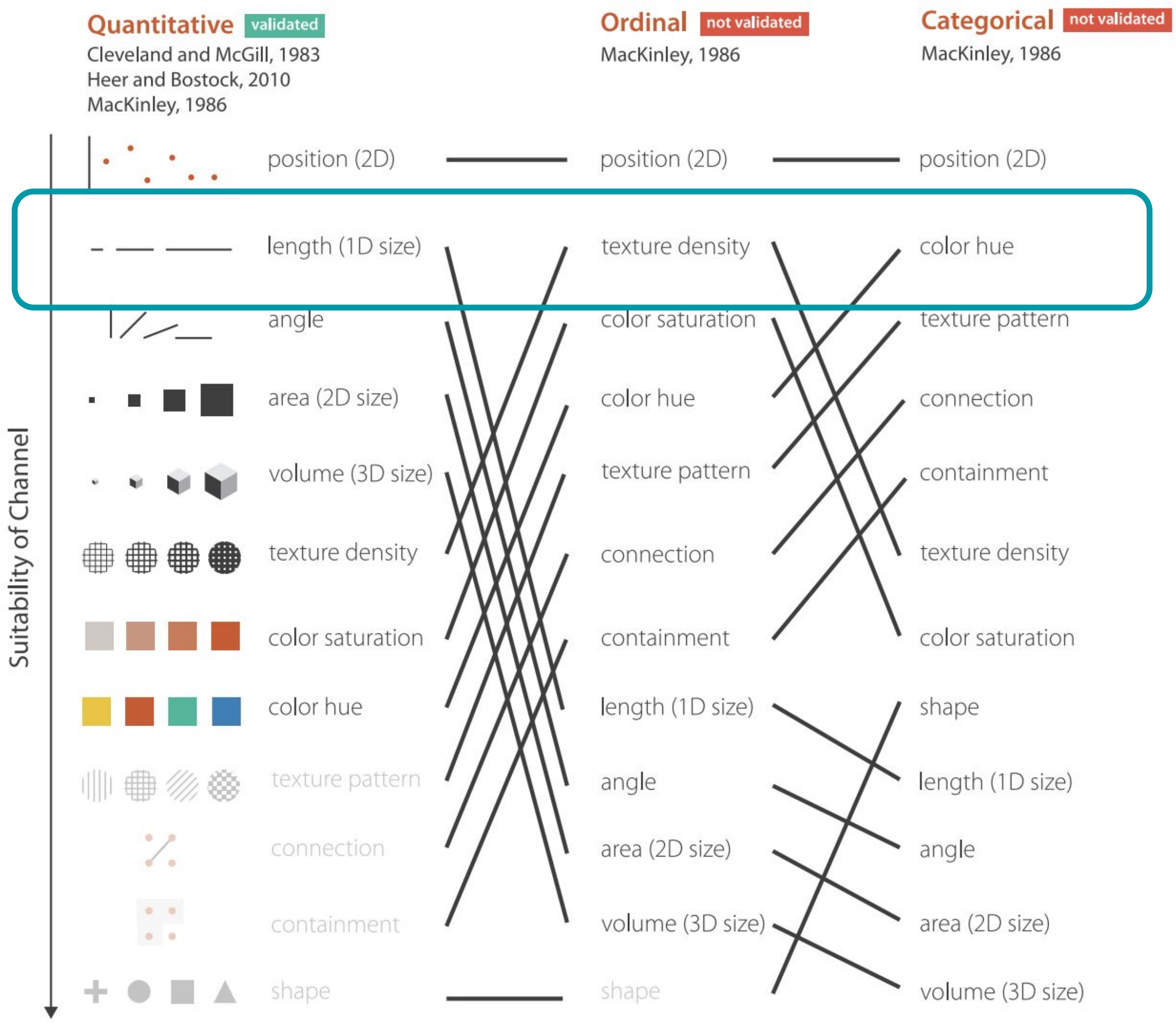


03

Best practices

Relative location (position) of elements using the same scale is the most effective property regardless of the data type

(maps, axes…)



03

Best practices

Length works better than angles (e.g. bar chart VS pie chart) when comparing numeric values.

Colors help to discriminate non quantitative values, as well as texture (gradation) for ordinal values.

What do you see in these charts?

02

Test

Answer the questionnaire (15 minutes)

https://forms.gle/s7mWD5mLtEyY6iL37

Thanks!

Do you have any questions?

marilena.daquino2@unibo.it

https://github.com/marilenadaquino/information_visualization